

# Spatial Analysis Report — Regional Sales & Underserved Markets

Dataset: india\_regional\_sales\_synthetic.csv. The analysis aggregates monthly city-level records using State, City, Postal Code and latitude/longitude, then combines demand, service capacity and user density to identify high-potential underserved markets.

## 1. Data Cleaning & Validation

Raw records: 300. Records used after required-field validation: 300. Location identifiers were trimmed and postal codes normalized to text. Numeric business and geographic fields were converted to numeric values; rows missing essential geography or analysis metrics were excluded.

## 2. Regional Aggregation

Metrics were aggregated at city level across the available monthly observations. Revenue and Users were summed; Demand and Service Capacity were averaged. User density is represented as average monthly users. Demand Gap is the positive difference between average demand and average service capacity.

## 3. High-Potential Underserved Regions

| Rank | City          | State          | Users | Revenue (₹) | Avg Demand | Avg Capacity | Demand Gap | Users/Month | Opportunity |
|------|---------------|----------------|-------|-------------|------------|--------------|------------|-------------|-------------|
|      | Visakhapatnam | Andhra Pradesh | 2,538 | 2,309,440   | 1.15       | 1.10         | 0.06       | 211.5       | 0.926       |
|      | Mysuru        | Karnataka      | 2,205 | 1,982,490   | 1.10       | 1.03         | 0.07       | 183.8       | 0.888       |
|      | Gurugram      | Haryana        | 2,169 | 2,009,689   | 1.13       | 1.06         | 0.07       | 180.8       | 0.880       |

## Interpretation

**Visakhapatnam, Andhra Pradesh:** 2,538 cumulative users and ₹2,309,440 cumulative revenue, with average demand 1.15 versus capacity 1.10. The positive demand-capacity gap of 0.06 indicates potential unmet demand under this dataset's scoring framework.

**Mysuru, Karnataka:** 2,205 cumulative users and ₹1,982,490 cumulative revenue, with average demand 1.10 versus capacity 1.03. The positive demand-capacity gap of 0.07 indicates potential unmet demand under this dataset's scoring framework.

**Gurugram, Haryana:** 2,169 cumulative users and ₹2,009,689 cumulative revenue, with average demand 1.13 versus capacity 1.06. The positive demand-capacity gap of 0.07 indicates potential unmet demand under this dataset's scoring framework.

## 4. Visualization & Submission

The accompanying interactive HTML map plots every analyzed city using latitude/longitude. Red markers identify the three highest opportunity scores; blue markers show the remaining locations. Clicking a marker displays users, revenue, demand, capacity and opportunity metrics.

**Important:** The dataset is synthetic, so these results should be presented as an analytical demonstration rather than real-world market demand evidence.